

Discrete~ Event Simulation Coursework: Hotel Maintenance

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Introduction

This report investigates the maintenance system of *Crymu Hotel & Spa*, using Simul8 to model the handling of scheduled and emergency maintenance tasks. Two entry points, *Scheduled* and *Emergency*, represent the two maintenance types entering the system. Jobs are allocated to a technician at the start of the day; however, emergency tasks interrupt the schedule and take precedence.

Model assumptions

The technician works weekdays only, with a 30-minute daily break, and no leave or holidays are taken. Scheduled tasks are assigned exactly at 9:00 AM. Any task not started by the end of the day is discarded and assumed to be rescheduled as a new task later. Tasks already in progress are held and resumed the next working day. The maintenance unit is assumed to operate without failure.

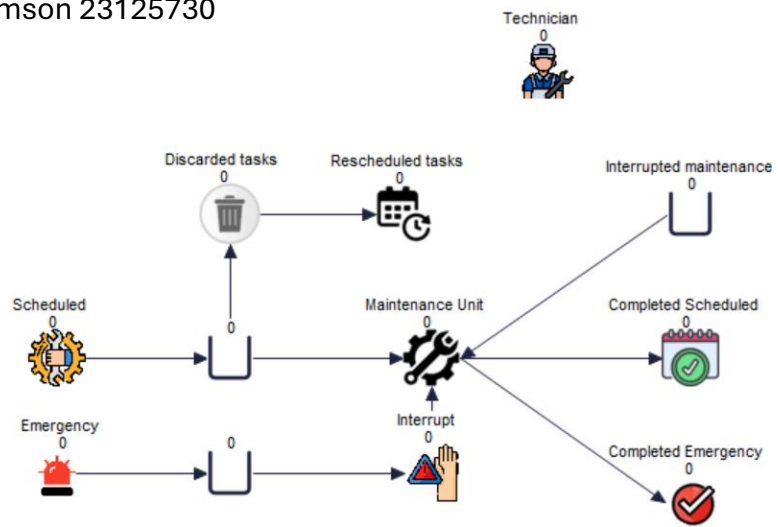
Clock

The simulation is set to operate 24/7 to reflect the hotel's continuous operation, allowing emergency repairs to enter the system outside the maintenance unit's working hours. I have selected hours as the time unit and set the result collection period to 26 weeks (6 months) to ensure realistic, long-term performance metrics.

Travel times are set to 0, as routing arrows are used solely for task direction and do not impact timing. A 336-hour (2-week) warm-up period is implemented to stabilise technician utilisation and emergency tasks, before collecting results. This period is not important for scheduled tasks as they are reset daily and do not require a warm-up to reach steady-state conditions. However, it is crucial for emergency tasks due to their unpredictable nature and potential to span overnight or over weekends.

Technician

I then created my technician resource. Only one technician is available, working a weekday day shift (09:00–17:00) with 100% assumed efficiency. Several methods for incorporating a 30-minute break were explored, including shift splitting. Ultimately, a break was implemented by adjusting the maintenance unit's efficiency using a time-absolute distribution triggering at 12:30 (± 5 minutes, modelled with a normal distribution) and a triangular distribution for break duration (28.2–31.8 minutes).



Warm Up Period

Start results collection after this number of hours:

336

☒ Include in displayed clock time

Results Collection Period

Time Format

☒ Time and Day
 ☐ Time only

Percent

☐ Percent
 ☒ Simple unit count

Preview

23:59

Sunday
Week 28

Properties

Additional

Graphics

Running Time

Daily start time (HH:MM)

00:00

Daily end time (HH:MM)

00:00

Duration of day (HH:MM)

24:00

Days per week

7

Default (Run-Spin)

Apply

Run for 24hrs:

Apply

Results Collection Period

Time Units

Seconds

Minutes

Hours

Days

Simple Unit Count

Decimals: 0

Description:

For units smaller than seconds use decimals of units e.g. 0.0001 = 1 millisecond

Length of Break Times

Timing (hours)

Lower: 0.47

Mode: 0.5 Upper: 0.53

Distribution: Triangular

OK Cancel Help Resources

Shift Availability

Shift:

- 1 Day Shift
- 0 Morning shift
- 0 Afternoon shift
- 0 weekday

Availability: I

OK

Cancel

Help

Shift Work Patterns

Shift Properties

Day Shift

Start Time: 09:00

End Time: 17:00

Days

☐ Every Day

☒ Weekly Pattern

☒ Mon ☐ Sat

☒ Tues ☐ Sun

☒ Wed

☒ Thur

☒ Fri

OK Cancel Help

Maintenance Types



In my model, I created two starting points; Scheduled and Emergency. The inter-arrival time for Scheduled is set to 09:00 daily via creating a time-absolute distribution with

fixed 0 variance. I then defined the batch size by a uniform distribution [1, 5], meaning 1 to 5 jobs are generated each day with equal likelihood. By applying a constraint and using a custom shift, *Weekday*, I ensure these tasks are only allocated Monday through Friday. As the technician is assumed to not work weekends, we also assume no maintenance will be scheduled. I then set the Emergency inter-arrival time to a

Batching

Batch size leaving
this object:

Lower Bound: 

1

Upper Bound:

5

Distribution:

Uniform 

Time Absolute Distribution	
jobs per day sched	
Times to be returned by this distribution:	Distribution for variability around this time:
9:00	Fixed 0

Shift Properties

weekday|

Start Time: 00:00

End Time: 24:00

Distribution: Poisson

Days

☐ Every Day

☒ Weekly Pattern

☒ Mon ☐ Sat

☒ Tues ☐ Sun

☒ Wed

☒ Thur

☒ Fri

Shift Properties

weekday

Start Time: 00:00

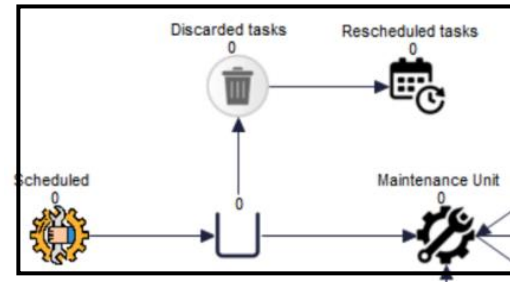
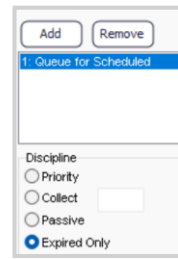
End Time: 24:00

Days

☐ Every Day

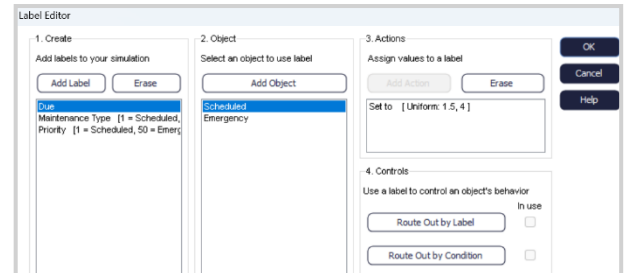
☒ Weekly Pattern

Each maintenance type feeds into a separate queue. Scheduled tasks are routed to the maintenance unit, which operates on the *Day Shift*. This is the same as the technician's schedule so maintenance can begin immediately as the jobs enter the system at 09:00. The scheduled queue has a shelf life of 8 hours to reflect the working day; any tasks not started by the end of the shift are directed into the Discarded tasks activity and removed from the system, allowing the queue to reset daily.

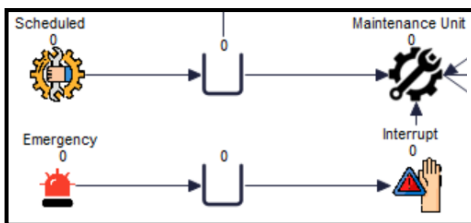


Labels

Labels play an essential role in my simulation model. Three labels were created: *Maintenance Type*, *Priority*, and *Due*. The *Maintenance Type* label, applied at the start points, assigns a fixed value of 1 to scheduled maintenance tasks and 2 to emergency tasks, enabling logical routing and task differentiation. The special label *Priority* was used to prioritise emergency maintenance by assigning a high fixed value of 50, while scheduled tasks were given a lower value of 1.



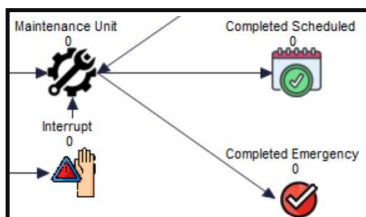
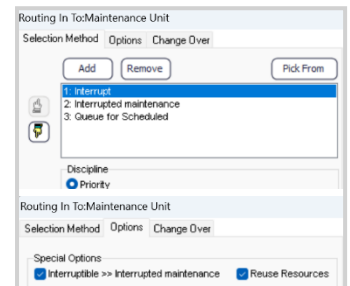
Finally, the *Due* label represents task duration, with scheduled tasks following a uniform distribution between 1.5 and 4 hours, and emergency tasks following an exponential distribution with a mean of 3.2 hours, as specified in the brief.



Emergency Maintenance (Interruption)

The Emergency queue routes into an *Interrupt* activity with a fixed duration of 0. This activity allows emergency maintenance to interrupt scheduled tasks using special labels. In the maintenance activity, I set the routing-in method to *Priority* and added the corresponding label in the Actions tab. I then set the timing to a fixed distribution and assigned

the *Due* label as the value, which records the remaining time of interrupted tasks, so they can resume where they left off. The activity was configured to be interruptible, with interrupted tasks held in the *Interrupted Maintenance* queue. The technician resource was also set to be reusable and allowed to *interrupt other work if necessary*. Thus allowing scheduled tasks to be paused and Emergency to start immediately during the maintenance unit operating hours.

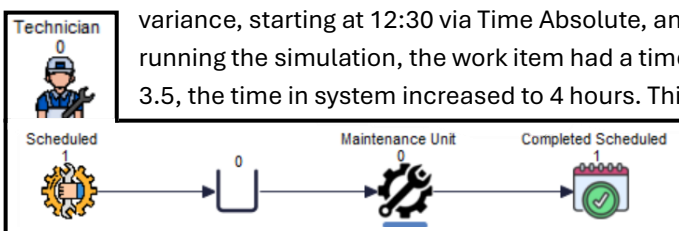


Maintenance Completion

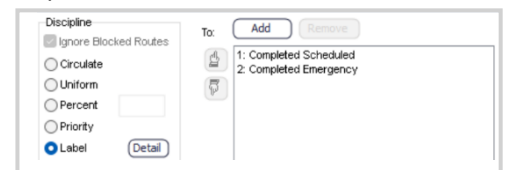
Once the Maintenance Unit Activity has successfully completed a task, the work item is routed out via the *Maintenance Type* label created prior, so we are able to differentiate the results for the types of maintenance throughout our simulation and once

Validation

To validate the break, I removed emergency maintenance, the warm-up period, and set the result collection to 1 day. I changed the scheduled maintenance batch to a fixed 1 distribution and adjusted the due label timing to fixed 3.49, ensuring the task ended just before 12:30. In the Maintenance activity efficiency, I set the time between repairs to fixed 0 variance, starting at 12:30 via Time Absolute, and configured the repair duration to fixed 0.5 (30 minutes). Upon running the simulation, the work item had a time in system of 3.49 hours. After adjusting the due label duration to 3.5, the time in system increased to 4 hours. This confirmed that the break occurred at 12:30 for 30 minutes, the work item was paused correctly, and the remaining time was accurately recorded.



To further validate, I ensured that: 1-5 scheduled jobs arrived daily at 09:00, not arriving on weekends, no work taking place in the maintenance unit outside of the *Day Shift*, Emergencies interrupt Scheduled, Scheduled work items wait in the interrupt queue, resuming after Emergency finishes and Emergencies can arrive at any time. I also calculated the percentages of maintenance job types in the system, technician utilisation and the expected time in the system, using probabilities and averages.

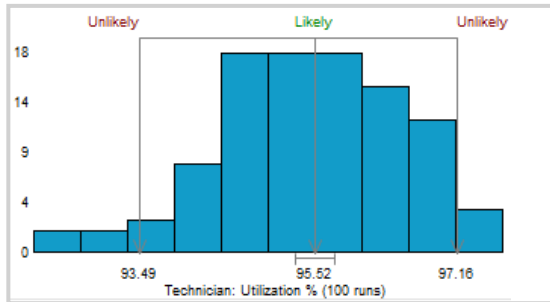


All of my results are very close to the simulation results, validated from running 100 trials of 26-week collection period and 2-week warm up period. My results below also use the same trials:

	Average	Simulated	Valid?
Maintenance (completed Scheduled)(%)	45.01	43.91	Yes
Maintenance (uncompleted Scheduled) (%)	35.06	36.17	Yes
Maintenance (completed Emergency) (%)	19.93	19.92	Yes
Uncompleted Emergency (%)	0.00	0.00	Yes
Time in system (Completed Scheduled)	20.31 hrs	20.46 hrs	Yes
Technician Utilisation (%)	94.10	95.56	Yes

Results

Technician Utilisation (95% CI) [95.26, 95.49, 95.71]



Metric (hours)	Min	Avg.	Max
Time scheduled rooms spend in the system	1.73	20.46	155.02
Number of scheduled rooms waiting for maintenance	0.00	0.57	5.00
Waiting time of scheduled rooms for service	0.00	5.83	8.00
Time unscheduled rooms spend in the system	0.14	26.24	98.50
Number of unscheduled rooms waiting for maintenance	0.00	0.35	1.00

“What If?” scenarios:

What if the technician works weekends? Keeping everything in the model the same except changing the *Day Shift* to *everyday* and not allocating a shift to constrain the Scheduled start points. Running the same as prior; 100 trials, 2-week warm-up period and 26-week collection period, we obtain the results:

	Weekdays Only	Works weekends	(%) change	Improvement?
Avg. time in system (Emergency) (hours)	26.24	12.54	-52.21	Yes
Max time in system (Emergency) (hours)	98.50	56.50	-42.64	Yes
Avg. time in system (Scheduled) (hours)	20.46	13.78	-32.66	Yes
Maintenance (completed Scheduled) (%)	43.91	54.64	+24.45	Yes
Technician Utilisation (%)	95.56	94.03	-1.60	No

These results demonstrate a strong maintenance optimisation potential by having the technician work weekends.

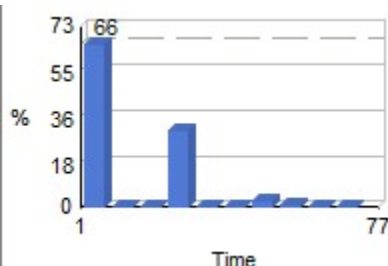
What if the arrival of emergency tasks change? To simulate this, I kept everything the same, however I changed the average of the Poisson distribution by increments of 20 hours determining the inter-arrival. Running 100 trials, 2-week warm-up period and 26-week collection period, we obtain these results, concluding a visible trend as the Poisson average increases, time in system and technician utilisation decrease.

	5	25	45	65
Avg. time in system (Emergency) (hours)	1648.91	33.68	26.24	24.19
Avg. time in system (Scheduled) (hours)	0.00	33.04	20.46	17.71
Technician Utilisation (%)	100.00	98.11	95.56	94.07

Summary and recommendations

The simulation of the *Cymru Hotel & Spa*'s maintenance system shows that emergency tasks effectively take priority over scheduled ones, but scheduled task completion is impacted by interruptions.

To further optimise the system, introduce a part-time technician to work 3 weekdays, with scheduled jobs adjusted to Uniform[3,5]. The current technician would independently cover the other two with Uniform[1,3] jobs. This maintains the weekly average (15 jobs) but distributes workload more effectively, although requires another employee. Also having a technician operating the maintenance unit over the weekend would be an effective way to optimise maintenance, demonstrated by the results from the “What if?” scenario, where 66% of the wait time is between 1- 8 hours for scheduled. The hotel could also consider acquiring an on-call technician for off-hours emergency repairs.



This model does have limitations: assuming a single technician with fixed hours and perfect efficiency, ignoring factors like absences, travel time, task rescheduling, task complexity, minimal categorisation, changing hotel occupancy and customer impact. These limitations point out areas where the model oversimplifies reality, making it less flexible and adaptable to varying conditions that occur in real-world operations.